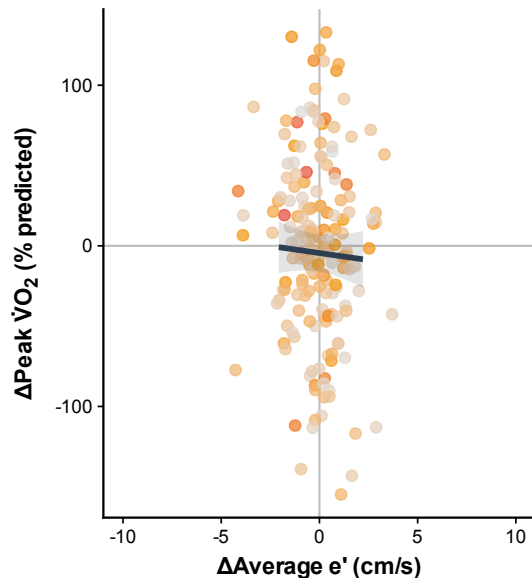


(A)

 Δ Average e' vs Δ Peak $\dot{V}O_2$

Mixed-model beta = -1.76, p = 0.548



(B)

 Δ Average e' vs $\Delta\dot{V}E/\dot{V}CO_2$

Mixed-model beta = 0.13, p = 0.616

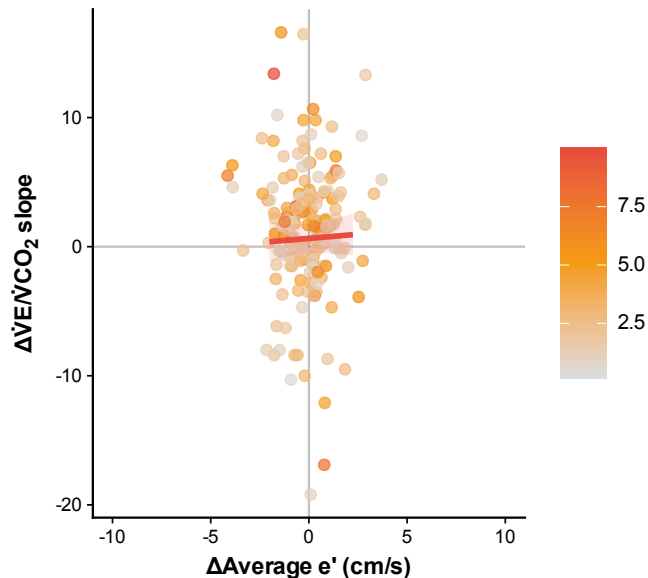


Figure S4. Exploratory within-patient longitudinal coupling between change in average mitral annular e' and change in cardiopulmonary performance. Points represent follow-up studies relative to the first paired observation with both e' and the given outcome available. Point color encodes years since the paired baseline; solid lines and shaded bands show the adjusted mixed-model association.